House Price Prediction - Project Summary

Objective

The objective of this project was to build a machine learning model capable of predicting house prices based on various property features such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and other amenities.

Dataset

The Housing Prices Dataset was obtained from Kaggle and contains information about residential properties and their corresponding prices. The dataset includes both numerical and categorical features related to house characteristics.

Data Preparation

The dataset was loaded using Pandas and examined for missing values and duplicate records. No missing values or duplicate rows were found in the dataset. Categorical variables such as main road access, air conditioning, furnishing status, and other yes/no features were converted into numerical format using one-hot encoding. The processed dataset was then prepared for model training and evaluation.

Models Used

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio.

Results

The performance of both models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

- Linear Regression: R^2 Score = 0.653
- Random Forest Regressor: R^2 Score = 0.612

Linear Regression performed better than Random Forest on this dataset and achieved moderate predictive accuracy.

Key Findings

Correlation analysis showed that area, bathrooms, air conditioning, stories, and parking were among the most influential factors affecting house prices. The analysis also indicated that unfurnished houses generally had lower prices compared to furnished houses.

Conclusion

The project demonstrated that machine learning can be used to estimate house prices based on property characteristics. The Linear Regression model was able to explain approximately 65% of the variation in house prices, making it useful for generating reasonable price estimates. Real estate businesses can use insights from this analysis to support property valuation and pricing decisions.